Supplement: Information-optimal illumination for dead-time-limited single-pixel imaging

[author block — user decision (mirrors main text)]

Numbering "Theorem 1–3" below refers to the main text; equations and propositions local to this supplement carry the S-prefix. All frozen constants trace to the R10/R11/R13/R14 rulings cited in the main-text source comments.

S1 The framework theorem: one master concave program

The framework theorem generalizes the main-text design program: with the random-hold kernel $J_H(\rho,\nu)$ of Theorem 2 inside the rank-one atoms $H_z = \nu J_H(\rho_z,\nu)\, q_z q_z^\top$, the design-measure problem remains concave, and every true interior optimum of the declared problem satisfies one generalized reduced-sensitivity/KKT system: the sensitivity $\psi(z) = M\,\mathrm{tr}[V^{-1}H_z] - \sum_m \beta_m c_m(z)$ obeys $\psi(z) \le \eta$ with equality on support atoms, and at any support atom interior in a smooth scalar coordinate y, $\partial_y \psi = 0$ with $\partial_y^2 \psi \le 0$. The load coordinate yields the information-water-filling equation — $M\nu J_H'(\rho)\,\ell$ equal to the marginal resource shadow price, with $\ell = q^\top V^{-1} q$ — while the support-scale and weight-power coordinates equate $2M\nu J_H(\rho)\, q_y^\top V^{-1} q$ to the corresponding marginal dose/peak/dispersion price: the apparent "conditioning penalty" is not an ad hoc term but the V^{-1} geometry inside the reduced sensitivity, which also explains why raising proxy contrast from $p = 1$ to $p = 2$ can lower the true objective. One caveat is kept deliberately: the master design problem places load, support scale, and intensity weighting in one constrained concave measure optimization, and its support atoms satisfy a common reduced-sensitivity condition; but the empirically observed occupancy ($k \approx 32$) and weighting ($p \approx 1$) optima *motivate* these coordinates and are not themselves claimed as continuous KKT theorems — they were selected from discrete grids and judged by image endpoints, not by the master $\log\det$ objective.

S2 The alignment proposition

At a fixed design iterate with directional values $\ell_i = q_i^\top V^{-1} q_i$, allocate scalar loads under the budget $\sum_i \pi_i \rho_i = B$ to maximize $F(\rho) = \sum_i \pi_i \ell_i J(\rho_i)$; on an interval where J is increasing and strictly concave, the water-filling optimum satisfies $\ell_i J'(\rho_i^*) = \beta$ at every interior coordinate. A single global source induces $\tilde\rho_i = \kappa u_i$, with $u_i = a_i^\top x$ the bucket brightness.

Proposition S1 (alignment and residual bound) *The fixed-flux allocation is exactly optimal for this scalar subproblem if and only if there exists β with*

$$\ell_i\, J'(\kappa u_i) = \beta \tag{S1}$$

for all interior active directions, with KKT-compatible inequalities at the bounds. If moreover $-\ell_i J''(\rho) \ge m > 0$ throughout the admitted interval, then, with residuals $r_i = \ell_i J'(\kappa u_i) - \beta$,

$$0 \;\le\; F(\rho^*) - F(\tilde\rho) \;\le\; \frac{1}{2m}\sum_i \pi_i\, r_i^2. \tag{S2}$$

A computable projected-KKT residual — not C_u alone — therefore certifies whether one global flux knob is close to the scalar water-filling optimum. In power-law form, if $J'(\rho) \asymp C\rho^{-s}$ and $\ell_i \asymp u_i^\alpha$, fixed flux aligns when $\alpha \approx s$; for the rising kernel $J = \rho/(1+\rho)$ the local slope $s(\rho) = 2\rho/(1+\rho)$ runs from 0 toward 2, so the often-invoked $\ell \propto u^2$ alignment is only a high-load approximation. Without the alignment condition there is no scene-independent constant-factor guarantee: two directions with $u_1/u_2 \to \infty$ but $\ell_1/\ell_2 \to 0$ drive the fixed-flux objective ratio to zero, so fixed flux can be arbitrarily bad. With random holds, J' vanishes at the jitter cap and is negative beyond it, so fixed flux can overdrive the brightest directions and explicit gain control becomes necessary. This explains both sides of the companion's dev evidence [1]: RIDGE-SCAT32 with one global knob is near-optimal exactly when its realized residuals are small (self-water-filling), while the uniform servo — which loses by up to ~ 7 dB on sparse-support scenes — fails because forcing equal ρ_i discards a naturally favorable alignment between u_i and ℓ_i.

S3 Design machinery in full

S3.1 Task subspace and numerical ridge

The estimable subspace is frozen at $r = 200$: B is the top-200 eigenbasis of the fixed comparator's information matrix under the same pre-scan, dwell, and resource convention, computed before the optimization and held fixed, with a fixed numerical ridge $\epsilon_0 = 10^{-9}\,\mathrm{tr}(B^\top V_{\mathrm{FIXED}^*} B)/r$ (R13 §6). A design-dependent ridge is not permitted: it would change the objective derivative and invalidate the sensitivity calculation.

S3.2 The near-optimality certificate

By concavity of $\log\det$ and affineness of $\xi \mapsto V(\xi)$, a feasible design ξ^* is optimal iff there exist constraint multipliers such that the reduced sensitivity $s(a) = \mathrm{tr}[V(\xi^*)^{-1}H(a)] - \sum_j \alpha_j e_j(a) - \sum_k \beta_k c_k(a)$ satisfies $s(a) \le \eta^*$ over $\mathcal{A}$ with equality on the support (the constrained Kiefer–Wolfowitz / general equivalence theorem for nonlinear models [2, 3]). The confirmatory certificate evaluates this gap for the *deployed* balanced SCAT32 design against the continuous optimum of the dose-safe class over the expanded dictionary $D_{\mathrm{cert}} = D_{\mathrm{load}} \cup D_{\mathrm{gain}}$ (§S3.4). With per-atom design value $d_a = \mathrm{tr}(V^{-1} M_{\mathrm{main}} H_a)$, one incident-budget multiplier $\theta \ge 0$, budget b, and upper/lower per-pixel-dose multipliers $\mu^+, \mu^- \ge 0$, the full dose-constrained dual is

$$G_{\mathrm{full}} = \min_{\theta,\mu^+,\mu^- \ge 0} \Big[\max_{a \in D_{\mathrm{cert}}} s_a(\theta,\mu^+,\mu^-) - d^\top \xi + \theta b \Big], \qquad G_{\mathrm{full}}/r \le 10^{-2}, \tag{S3}$$

where $s_a(\theta,\mu^+,\mu^-) = d_a - \theta\rho_a - \sum_j (\mu_j^+ - \mu_j^-)\,\partial d_j/\partial \xi_a$ is the dose- and budget-priced reduced sensitivity, the inner maximum uses coverage-seeded column generation and finishes with an exact scan over all of D_{cert} [4], and every upper and lower pixel-dose inequality carries its own multiplier (R15 / R17 §4.4). The bar $G_{\mathrm{full}}/r \le 10^{-2}$ certifies local D-efficiency of at least $\exp(-0.01) = 0.99005$ relative to the continuous optimum of the expanded dose-safe class. A-risk and spectral constraints are deliberately excluded from the dual target class, making the certified class larger and the result stronger; the deployed SCAT32 rows still satisfy the A-risk, spectral, peak, and ceiling disclosures separately. Retained numerics: active-dose toy agreement $\le 10^{-8}$, primal dose/budget feasibility, the full-dictionary scan residual, dual/complementarity reporting, and the finite μ cap as a conservative restriction disclosed as `MU_CAP_ACTIVE`. Continuous weights are rounded to an exact 972-row design by efficient rounding followed by deterministic exchange [5, 6, 7], admissible only if the exact D-efficiency ≥ 0.95.

S3.3 Guards as constraints

The raw optimum is admissible only after all frozen guards pass; each is a constraint of the master program or a post-rounding check, not a tunable proxy. *Peak (two-part, R13 A2):* a shape guard bounds unit-load support weights to $\frac{1}{4} \le w_j \le 4$ with support size $k \ge 16$, and a physical guard caps $\max_{i,j} a_{ij}^{\mathrm{physical}} \le 1536$ (the peak of a uniform $k = 16$ atom at the absolute load cap $\rho = 24$), so matched-intensity ridge atoms cannot silently demand more peak power. *Weight power (R10 G4):* the dictionary contains powers $p \in \{0, 1\}$ only; $p = 2$ is excluded because dev evidence showed higher proxy contrast can worsen conditioning and endpoint quality. *Per-pixel dose (R10 G5):* each pixel's mean exposure stays within $\pm 5\%$ of the image mean, enforced in-optimizer and re-verified after rounding; a necessary row bound follows, $\rho_r \le 1.05\, M\bar{d}/\max_j g_{rj}$ (R13 A5). *Spectral floor (R10 G6):* $V_{\mathrm{OED}} \succeq 0.5\, V_{\mathrm{FIXED}^*}$ on B, with a convex-mixture fallback whose coefficient is fixed from information matrices before any image endpoint is seen. *Kernel and trust (R11 G7–G8, R13 A1):* every

(ν, ρ) node is certified for exact-PMF mass, score, and derivative agreement, mean-information efficiency $J_{\text{mean}}/J_{\text{exact}} \geq 0.90$, finite-window RQL log-load bias $|\log(\rho^{\dagger}/\rho)| \leq 0.05$, ceiling probability $p_{\text{ceil}}(\rho, \nu) \leq 0.05$ per atom and ≤ 0.01 design-weighted. *A-risk (R10 G7):* $\operatorname{tr}(V_{\text{exact}}^{-1}) \leq 1.05 \operatorname{tr}(V_{\text{FIXED}^*}^{-1})$. Any violation is a declared failure state (`DICTIONARY_RANK_FAIL`, `CONTINUOUS_CERTIFICATE_FAIL`, `SPECTRAL_GUARD_FAIL`, `A_RISK_GUARD_FAIL`, `DOSE_GUARD_FAIL`, `ROUNDING_FAIL`), whose predeclared fallback is the fixed comparator; no threshold is retuned after freeze.

S3.4 The frozen dictionary

The exact searchable base dictionary D_{load} includes all cyclic translates of: scattered balanced supports at $k = 16$ and $k = 32$; the solid 4×4 patch; the committed LBLOB6×6 16-pixel support; bars/rectangles 1×16, 2×8, 4×4, 8×2, 16×1; compact 32-pixel square/rectangle variants; and the frozen 16-pixel ring/annulus family — with $p = 0$ and guarded $p = 1$ weights for every applicable support. The implementation supports variable k; all translation families use the circulant/FFT evaluator, and the dictionary manifest carries every family, translation, power, support size, and SHA256. These are the target-load-normalized atoms: each pattern's amplitude is chosen to hit a predicted load.

For the near-optimality certificate only, an additional gain-coupled family D_{gain} is formed: for every frozen base geometry and weight pattern, physical rows $a_{g,\gamma} = \gamma\, a_g^{\text{base}}$ with $\gamma \in \{0.2, 0.5, 1, 2, 5\}$, the base a_g^{base} normalized by the scene-independent physical convention of the fixed patterns, and predicted load allowed to *emerge* as $\rho_{g,\gamma} = \bar{\rho}\, \gamma\, (a_g^{\text{base}})^{\top} \hat{x}$ — never rescaled to a target load, so fixed-flux self-water-filling and bounded per-pattern gain control lie inside the comparison class. The same support, weight, physical-peak, ceiling, and RQL-trust admission guards apply to both parts; $D_{\text{cert}} = D_{\text{load}} \cup D_{\text{gain}}$ is SHA256-frozen before any confirmatory scene is opened.

S3.5 Budget accounting

For every arm both the incident budget $B_{\text{inc}} = \sum_i \lambda_{s,i} T$ and the detected budgets $B_{\text{det}}^{\text{pred}} = \sum_i \mathbb{E}[N_i]$, $B_{\text{det}}^{\text{obs}} = \sum_i N_i$ are reported, together with total and maximum per-pixel dose. B_{det} may serve as an electronics-throughput guard if a hardware limit is frozen before confirmation, but it is never the sole fairness constraint: for a nonparalyzable detector $\mathbb{E}[N \mid \rho] \simeq \nu\rho/(1 + \rho)$, whose marginal cost tends to zero at high load, so a detected-count-only budget can permit arbitrarily large incident power while appearing nearly cost-free.

S4 Monte-Carlo verification detail

The refined sweep (`jitter_sfi_v2`: lognormal holds, 100,000 frames per cell, common random numbers, 25-point log grid of spacing 1.157, $\nu \in \{200, 2000\}$) confirmed the R14 law out-of-sample. At $\nu = 2000$ the measured peaks are $\rho_* = 22.297$, 10.739, 5.173, 3.862, 2.153, and 1.607 at $c_v = 0$, 0.02, 0.05, 0.1, 0.2, and 0.3, respectively; both predictions registered before the data existed hit — $c_v = 0.02$ predicted ≈ 10.4, measured 10.739 (3%); $c_v = 0.2$ predicted ≈ 2.30, measured 2.153 (6%) — and every in-sample peak lies within one grid notch of its exact-law value (commits `8a627bb`, predictions registered pre-data; `1d8d7aa`, verdict). The $\nu = 200$ column is consistent with the crossover correction: the $c_v = 0$ peak at 9.28 tracks the exact ridge 10.04, and $c_v = 0.3$ measures 1.86 against the predicted 1.77.

A continuous-peak zoom (quadratic interpolation, local run, 150,000 frames) and an independent replication (Colab, 400,000 frames) then measured the four jittered peaks twice each at $\nu = 2000$: local 9.645, 5.154, 3.752, 2.106 and replicated 9.610, 5.700, 3.399, 2.051 at $c_v = 0.02$, 0.05, 0.1, 0.2, respectively. The pooled eight-measurement log-log fit gives slope -0.658 vs. the predicted $-2/3$ (1.3%); the constant sits about 5% below $2^{-1/3}$, at the edge of the run-to-run noise, so the frozen statement is: exponent confirmed at 1.3%, constant consistent within 5–10%. The end points replicate a mild $\approx -8\%$ low bias across both runs, left as an open higher-order note (candidate: a lognormal third-moment correction to the leading finite-variance law). One intermediate hypothesis is on the record: after the first two zoom points, an effective jitter-cost multiplier $k = 4/3$ was preregistered (commit `5e51088`) and was refuted by the very next measured point, whose deviation flipped sign (commit `1a32758`) — peak-location noise on flat information tops, not a systematic constant. The fuller three-hold-family exponent battery frozen in the ruling (tests E1–E5, with matched-CV gamma and bounded two-point holds and cross-fitted conditional-score estimation) remains a preregistered extension beyond this verdict.

S5 Secondary-arm and certificate full results

S5.1 Ridge operating-point and time-to-quality

[M1: per-dwell fixed-dwell gains of RIDGE-SCAT32 over SCAT32-060 and the safe reference, realized loads, requested vs. achieved mean load, `RIDGE_GUARD_CLIPPED` flags, ceiling fractions, and the nine-dwell S_{gate} curves for `RIDGE_SPEED_PASS`.]

S5.2 Near-optimality certificate distribution

[M1: per-cell full dose-constrained dual G_{full}/r over the 480 certificate cells, primal feasibility, exact D-efficiency, dual/complementarity, and `MU_CAP_ACTIVE` incidence, for `DOSE_SAFE_CERT_PASS`.]

S5.3 Context arms and design diagnostics

[M1: SCAT16/LBLOB16 context ladder, the dose-relaxed and dose-constrained OED design diagnostics, and the development-cell guard-versus-α collapse (`PATH_FEASIBLE_ALPHA`) — all no-endpoint.]

S6 Proofs

[supplement proofs at manuscript rigor: Theorems 1–3 (missing-information identity via the martingale/law-of-total-variance argument; long-window rate via the renewal local normal approximation; cubic optimum and small-jitter expansion) and Proposition S1; per the R14 ruling these are provable immediately, while the uniform two-parameter crossover requires a dedicated triangular-array renewal local-limit argument or retains its prediction label.]

References

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[7] Guillaume Sagnol and Radoslav Harman. Computing exact D-optimal designs by mixed integer second-order cone programming. *The Annals of Statistics*, 43(5):2198–2224, 2015. arXiv:1307.4953.